

# Homework Assignment 1

ANALOGUE INTEGRATED CIRCUIT DESIGN

ET4252

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| Source | Accuracy | Noise | $r_{\text{out}}$ | Total score (/15) |
|--------|----------|-------|------------------|-------------------|
| (a)    | 2        | 3     | 2                | 7                 |
| (b)    | 4        | 3     | 2                | 9                 |
| (c)    | 4        | 3     | 4                | 11                |
| (d)    | 4        | 5     | 5                | 14                |
| (e)    | 5        | 5     | 5                | 15                |

## Circuit A

### 1) Accuracy

$$\begin{aligned}
 V_B &= V_{CC} - V_{EB} \\
 I_B &= \frac{V_B}{R_B} = \frac{V_{CC} - V_{EB}}{R_B} \\
 I_{\text{out}} = I_C &\approx \beta I_B = \beta \frac{V_{CC} - V_{EB}}{R_B}
 \end{aligned}$$

Temperature dependence dominates, so the **\*\*most\*\*** temperature-dependent term is  $\beta$ .  $I_{\text{out}} \approx \beta \frac{(V_{CC} - V_{EB})}{R_B}$

**Grade 2**

### 2) Noise

Collector current noise:

$$i_{n,\text{collector}}^2 = 2qI_C$$

Noise due to resistor at o/p:

$$I_{n,\text{out}} = \beta^2 \frac{4kT}{R}$$

Total noise approximately:

$$i_{n,\text{total}} \approx \sqrt{(\sqrt{2qI_C})^2 + (\beta\sqrt{4kT/R})^2}$$

**Grade 3**

### 3) $R_{\text{out}}$

1.  $V_E$  at collector. At AC (emitter is ground),  $V_E = 0$ .
2. So no connection to collector,  $v_b = 0$ .

$$\frac{V_b}{R_B} + \frac{V_b}{R_\pi} = 0$$

Thus  $i_e = 0$ ,  $i_b = 0$ , so only the output resistance remains:

$$r_o = \frac{v_x}{i_x} = \frac{V_A}{I_C}$$

Base resistance does **\*\*not\*\*** boost  $R_{\text{out}}$ . No degeneration at emitter,  $V_{BE} = 0$ .

**Grade 2**

## Circuit B

### 1) Accuracy

Since the diode-connected transistor (Q1) has base and collector connected, Q2 copies the current (current mirror).

$$\begin{aligned} V_{CC} &= V_E \\ V_B &= V_{CC} - V_{EB} \\ V_{EB} &= V_E - V_B \\ I_R &= \frac{V_B}{R_C} = \frac{V_{CC} - V_{EB}}{R_C} \end{aligned}$$

KCL at Q1 base node:

$$I_R = I_{C1} + I_{B1} + I_{B2}$$

With matched devices:

$$I_{C1} = I_{C2} = I_C, \quad I_B = \frac{I_C}{\beta}$$

So:

$$I_R = I_C + \frac{2I_C}{\beta} = I_C \left( 1 + \frac{2}{\beta} \right)$$

Hence:

$$I_{\text{out}} = I_{C2} = I_C = \frac{V_{CC} - V_{EB}}{R_C \left( 1 + \frac{2}{\beta} \right)}$$

For  $\beta \gg 2$ :

$$I_{\text{out}} \approx \frac{V_{CC} - V_{EB}}{R_C}$$

**Grade 4**

## 2) Noise

3 noise sources contribute here.

Q2 output transistor (collector shot noise):

$$i_{n2,\text{shot}}^2 = 2qI_C$$

Q1 reference transistor (shot noise):

$$i_{n1,\text{shot}}^2 = 2qI_C$$

Diode-connected small-signal impedance is small:

$$Z_{Q1} \approx \frac{1}{g_{m1}}$$

So most of the noise appears over  $\frac{1}{g_{m1}}$  and is copied to the output (matched devices  $g_{m1} = g_{m2}$ ):

$$i_{n1,\text{out}} = g_{m2}v_{n,\text{base}} \approx \frac{g_{m2}}{g_{m1}} i_{n1} \approx i_{n1}$$

$R_C$  noise (Norton model):

$$i_{n,R_C} = \sqrt{\frac{4kT}{R_C}}$$

Total noise approximately (RSS):

$$i_{n,\text{total}} \approx \sqrt{(\sqrt{2qI_C})^2 + (\sqrt{2qI_C})^2 + \left(\sqrt{\frac{4kT}{R_C}}\right)^2}$$

$$i_{n,\text{total}} \approx \sqrt{4qI_C + \frac{4kT}{R_C}}$$

**Grade 3**

## 3) $R_{\text{out}}$

AC analysis, emitter = 0.

KCL at base node:

$$v_b \left( g_{m1} + \frac{1}{R} + \frac{1}{r_\pi} \right) = 0 \Rightarrow v_b = 0$$

So  $v_{BE2} = 0$ , hence  $g_m v_{BE} = 0$ .

So at the controlled source again:

$$i_x = \frac{v_x}{r_o}$$

Therefore:

$$R_{\text{out}} = \frac{v_x}{i_x} = r_o = \frac{V_A}{I_C}$$

**Grade 2**

## Circuit C

### 1) Accuracy

$$\begin{aligned}I_E &= \beta I_B \\I_B &= \frac{V_{CC} - V_{EB}}{R_B} \\I_B &= \frac{I_E R_E + V_{\text{REF}}}{R_B} = \frac{\beta I_B R_E + V_{\text{REF}}}{R_B} \\(R_B - \beta R_E) I_B &= V_{\text{REF}} \\I_B &= \frac{V_{\text{REF}}}{R_B - \beta R_E}\end{aligned}$$

For  $R_B = 0$ :

$$I_B = \frac{V_{\text{REF}}}{\beta R_E}$$

$$I_C = \beta I_B + I_B = (\beta + 1) I_B = (\beta + 1) \frac{V_{\text{REF}}}{\beta R_E}$$

$$I_{\text{out}} = I_C = \frac{V_{\text{REF}}}{R_E} \left( \frac{\beta + 1}{\beta} \right) \quad [\text{independent of } \beta \text{ for } \beta \gg 1]$$

**Grade 4**

### 2) Noise

$$i_{n,\text{total}} \approx \sqrt{(\sqrt{2qI_C})^2 + \left(\sqrt{\frac{4kT}{R_E}}\right)^2} = \sqrt{2qI_C + \frac{4kT}{R_E}}$$

**Grade 3**

### 3) $R_{\text{out}}$

$$i_b = \frac{v_e}{r_\pi}, \quad i_e = \frac{v_e}{R_E}$$

$$i_c = i_b + i_e = v_e \left( \frac{1}{R_E} + \frac{1}{r_\pi} \right)$$

$$i_c = \frac{v_c - v_e}{r_o} + g_m v_e = \frac{v_c}{r_o} - v_e \left( \frac{1}{r_o} - g_m \right)$$

Assume  $r_o \gg (R_E \parallel r_\pi)$ , so  $v_e = i_c(R_E \parallel r_\pi)$ .

$$i_c = \frac{v_c}{r_o} + i_c(R_E \parallel r_\pi)g_m$$

$$i_c(1 - (R_E \parallel r_\pi)g_m) = \frac{v_c}{r_o}$$

$$R_{\text{out}} = \frac{v_c}{i_c} = r_o(1 - (R_E \parallel r_\pi)g_m)$$

**Grade 4**

## Circuit D

Nullor implementation

### 1) Accuracy

$$I = \frac{V_{DD} - V_{\text{REF}}}{R}$$

**Grade 4**

### 2) Noise

Noise comes from the resistor only:

$$i_{n,R} = \sqrt{\frac{4kT}{R}}$$

So:

$$i_{n,\text{out}} = i_{n,R}$$

**Grade 5**

### 3) $R_{\text{out}}$

Ideal current source:

$$R_{\text{out}} = \infty$$

**Grade 5**

## Circuit E

### 1) Accuracy

$$I_{\text{out}} = \frac{V}{R + R_{\text{load}}}$$

For  $R \gg R_{\text{load}}$ :

$$I_{\text{out}} \approx \frac{V}{R} = 1 \text{ mA}$$

Load regulation error:

$$\frac{\Delta I_{\text{out}}}{I_{\text{out}}} = -\frac{R_{\text{load}}}{R + R_{\text{load}}} \approx -\frac{R_{\text{load}}}{R}$$

No special parameters, no temp dependent junctions, transistor param spread  $> R$  spread.

**Grade 5**

### 2) Noise

Resistor is the only noise source (ideal voltage source has no noise):

$$i_{n,R} = \sqrt{\frac{4kT}{R}}$$

**Grade 5**

### 3) $R_{\text{out}}$

$$R_{\text{out}} = R = 1 \text{ G}\Omega$$

Independent of load voltage, current constant regardless of load, basically ideal current source.

**Grade 5**

## A Handwritten Derivations

The following pages contain the original handwritten analyses for each current source circuit, covering accuracy, noise, and output resistance derivations.

## A.1 Circuit 1 – Single PNP Transistor Current Source

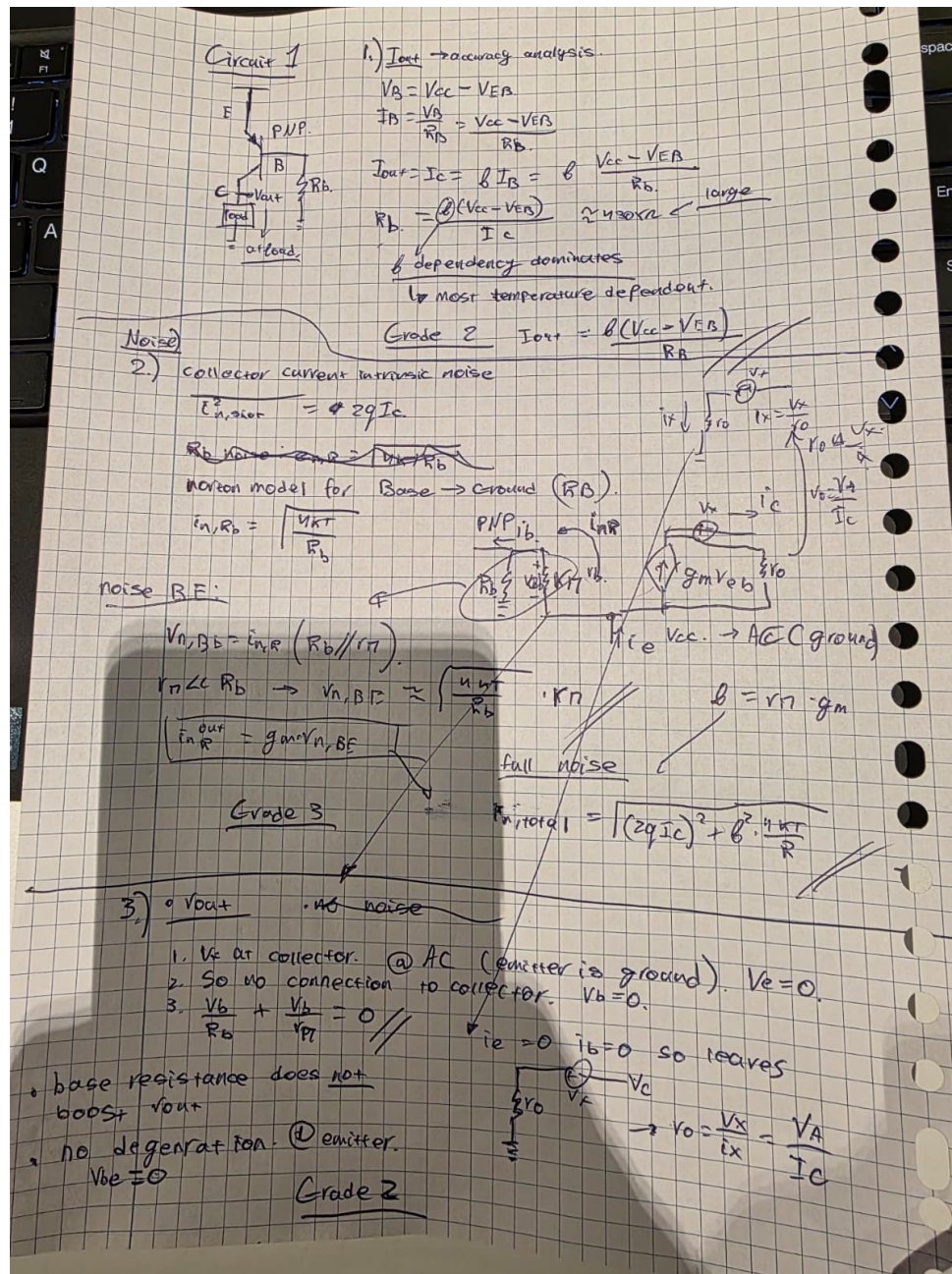


Figure 1: Circuit 1: Accuracy, noise, and output resistance analysis for the single PNP transistor current source.

## A.2 Circuit 2 – PNP Current Mirror

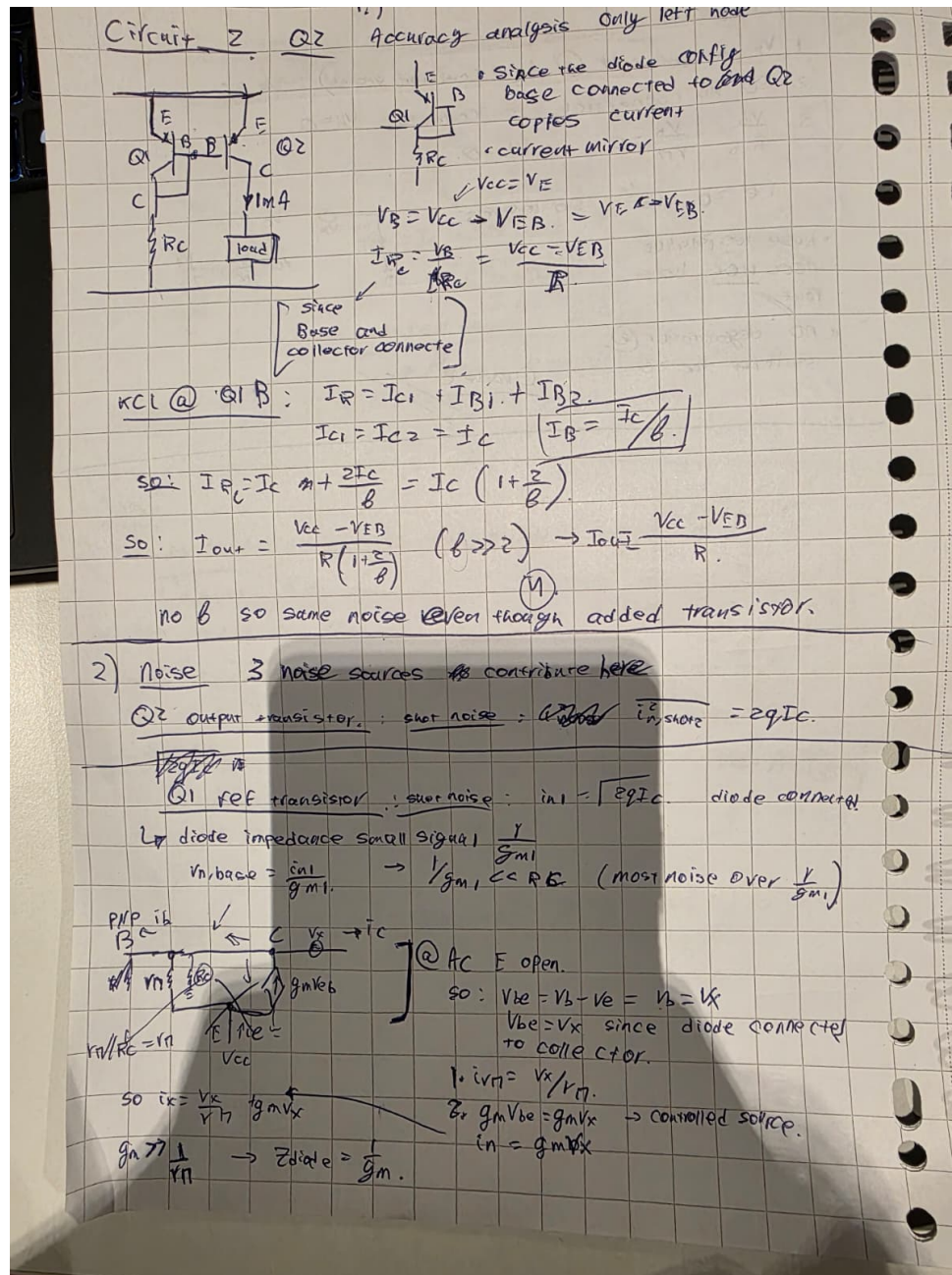


Figure 2: Circuit 2 (part 1): Accuracy and noise analysis for the PNP current mirror.

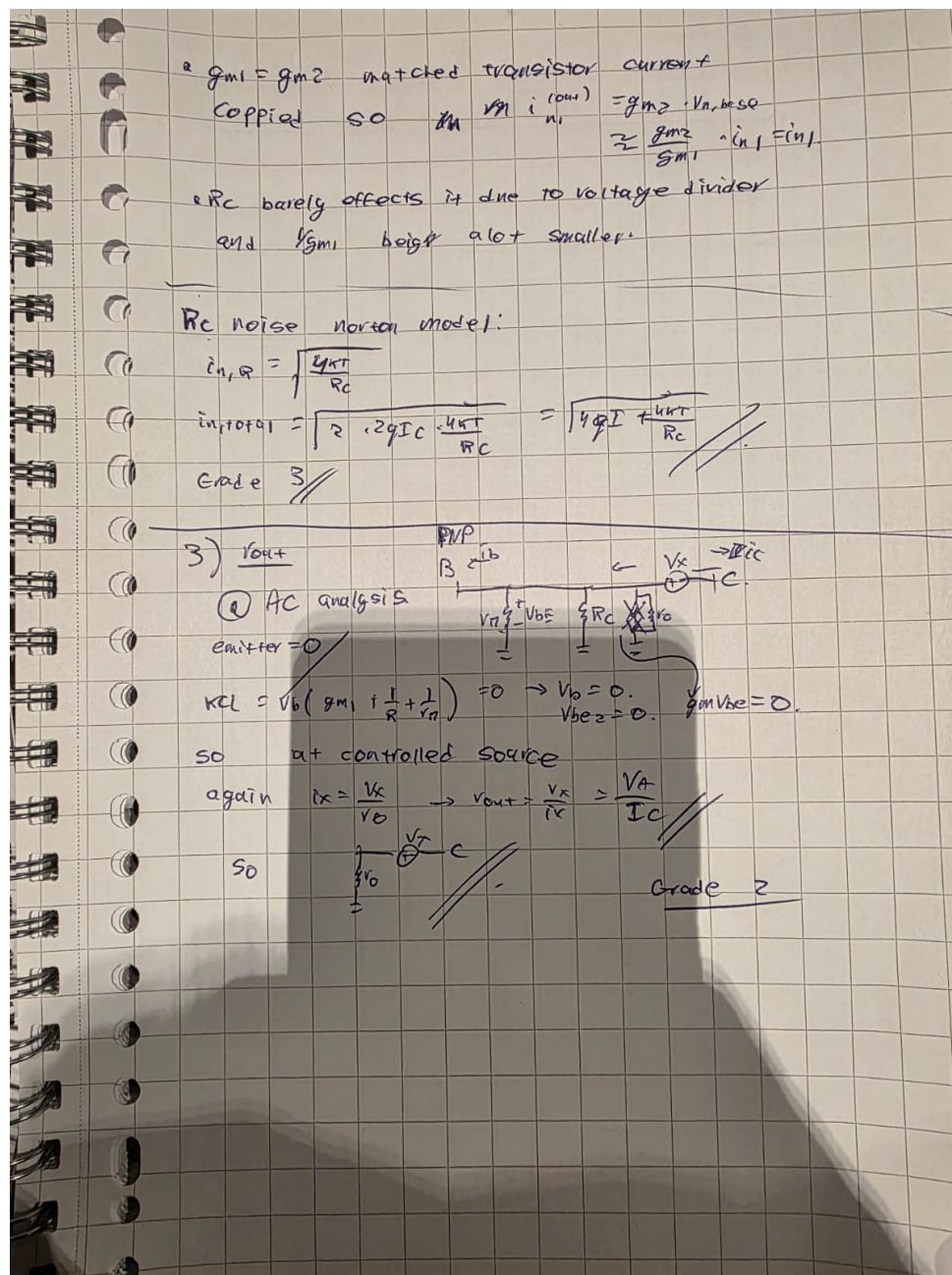


Figure 3: Circuit 2 (part 2): Matched transistor analysis,  $R_C$  noise contribution, and output resistance derivation.

### A.3 Circuit 3 – PNP with Emitter Degeneration Resistor

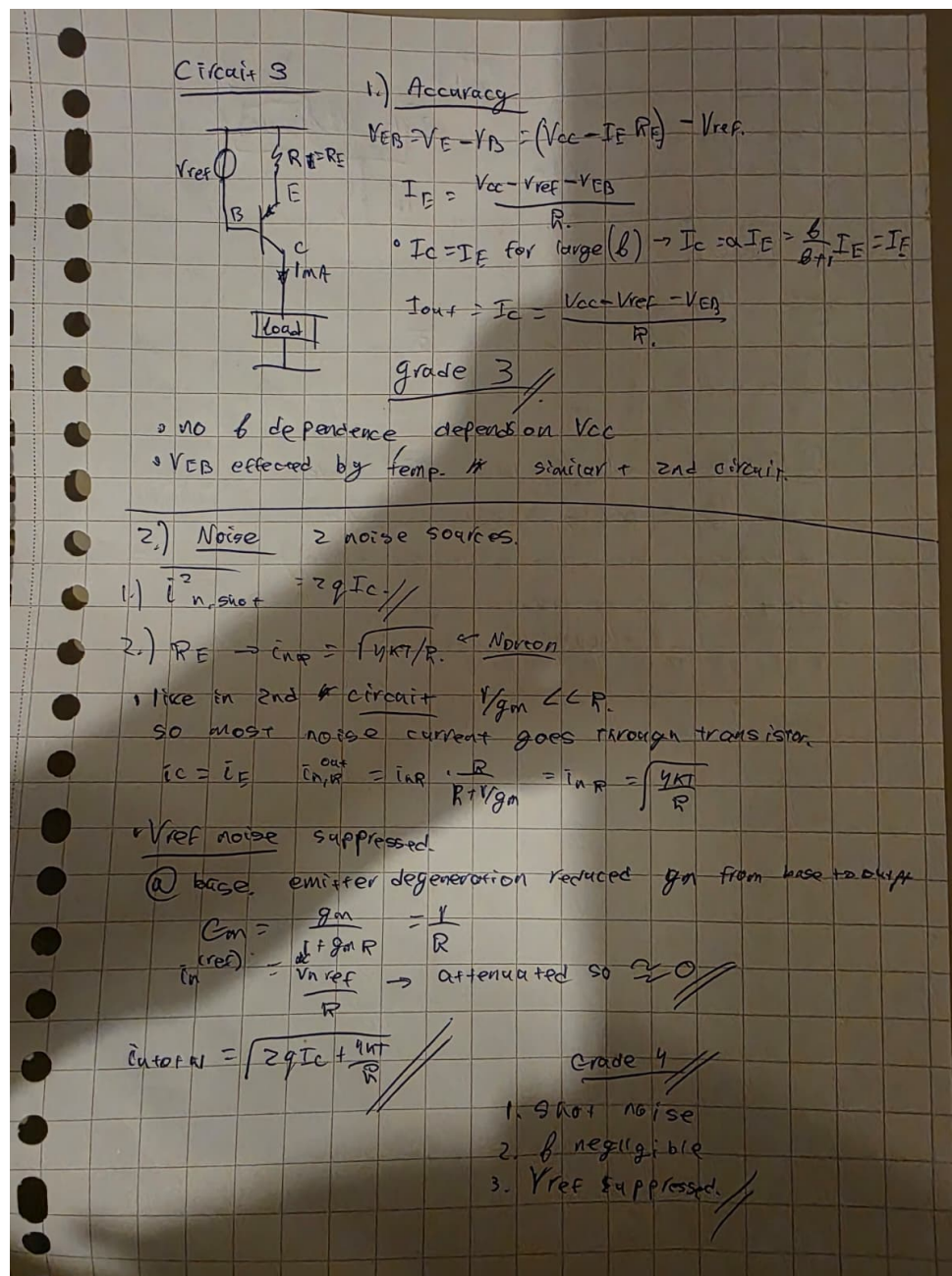


Figure 4: Circuit 3: Accuracy, noise, and output resistance analysis for the PNP current source with emitter degeneration.

#### A.4 Circuit 4 – Op-Amp Feedback Current Source

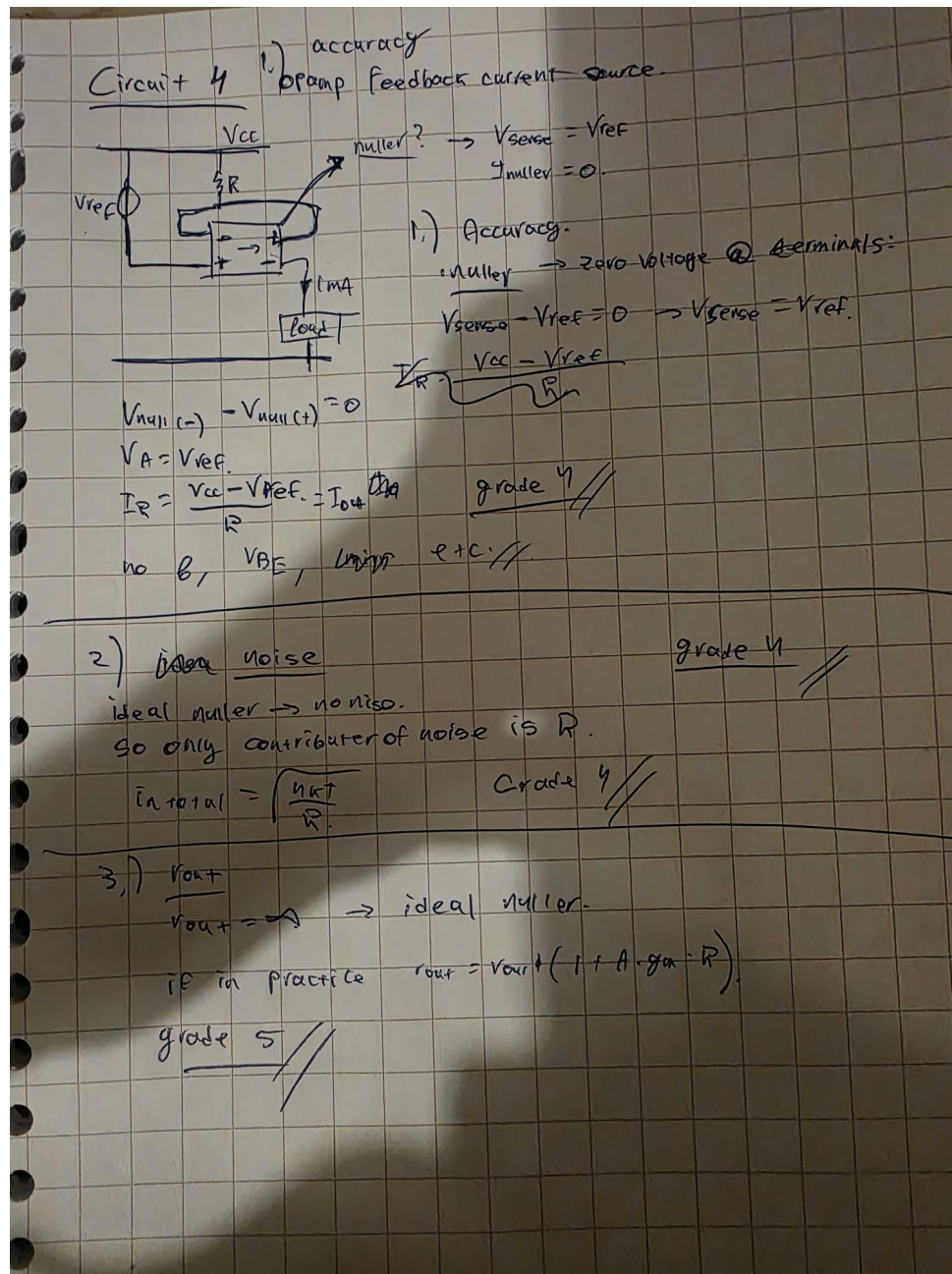


Figure 5: Circuit 4 (part 1): Accuracy, noise (ideal nuller), and output resistance analysis for the op-amp feedback current source.

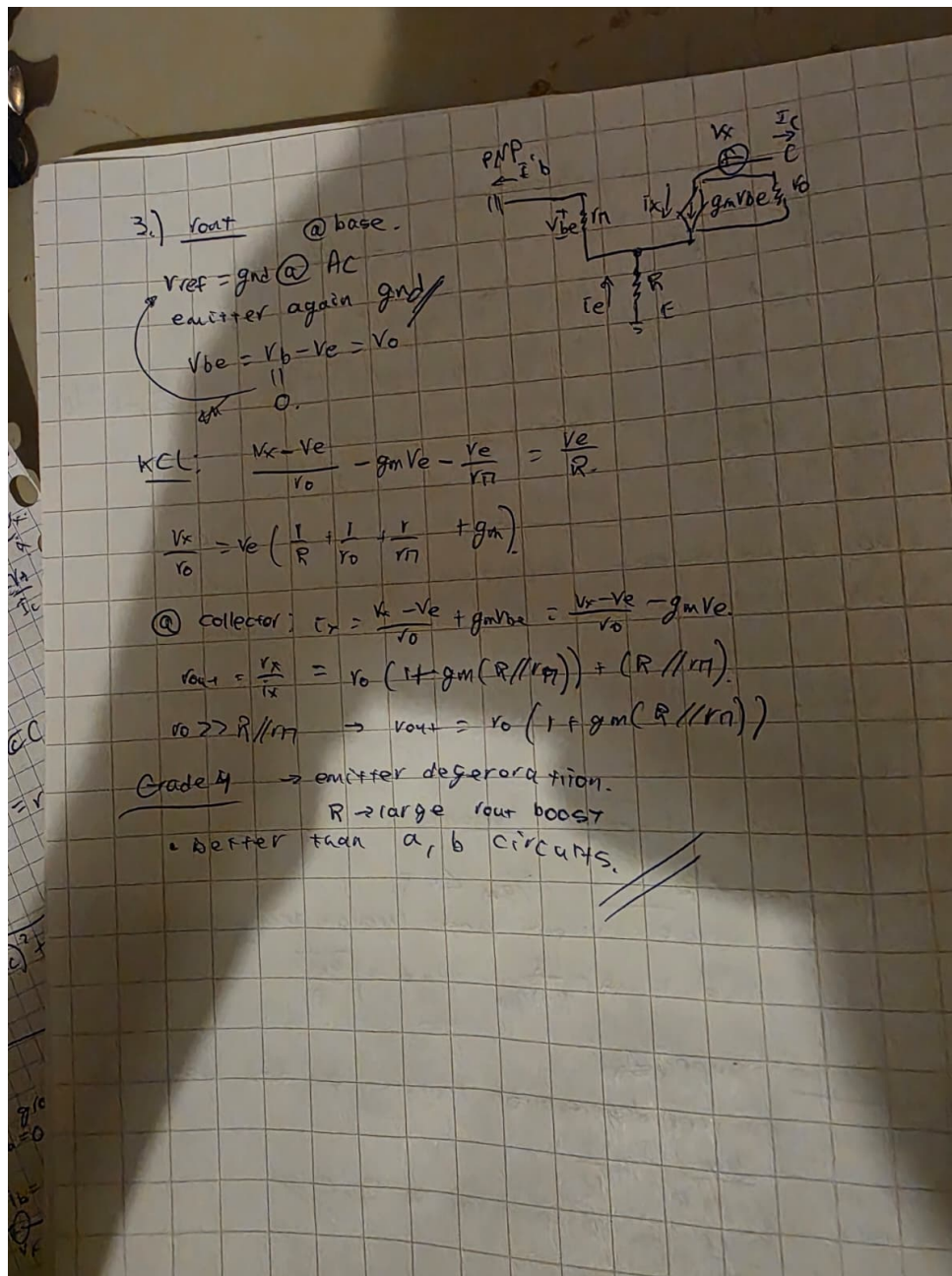


Figure 6: Circuit 4 (part 2): Detailed  $r_{out}$  derivation using PNP small-signal model with emitter degeneration.

## A.5 Circuit 5 – Voltage Source with Large Resistor

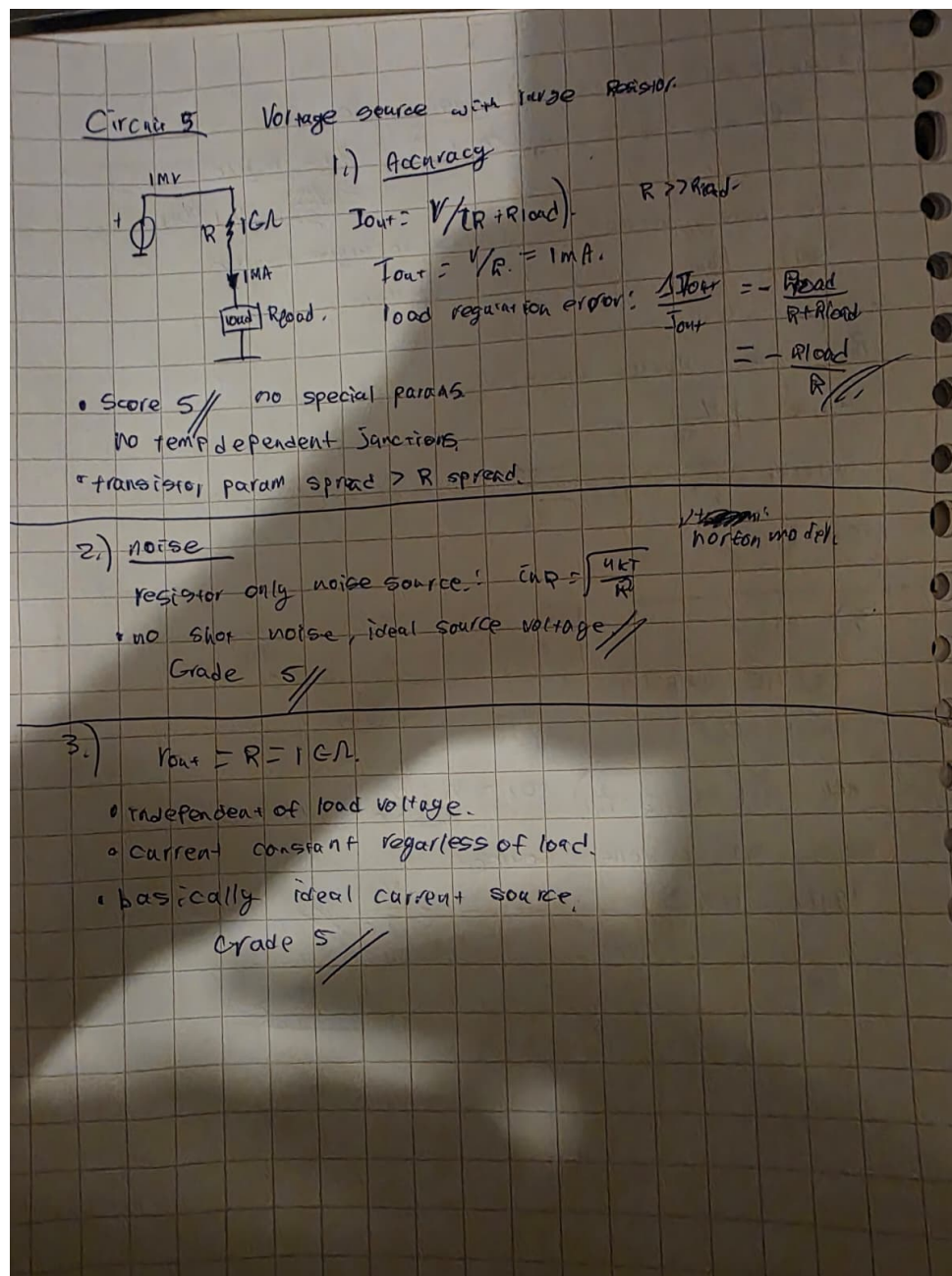


Figure 7: Circuit 5: Accuracy, noise, and output resistance analysis for the voltage source with large series resistor.